genetically modified crops are herbicide tolerant – Basta, Dhara mustard, insects resistance – Bt crops, flavrSavr – Tomato, Golden rice. Biopolymers are polyhydroxybutyrate (PHB), polylactic acid (PLA) and green fluorescent protein (GFP) is used to make biosensors. Other applications are biopharming, bioprospecting, biomedication and biofuel, etc.

Evaluation



1. Restriction enzymes are
   a. Not always required in genetic engineering
   b. Essential tools in genetic engineering
   c. Nucleases that cleave DNA at specific sites
   d. both b and c

2. Plasmids are
   a. circular protein molecules
   b. required by bacteria
   c. tiny bacteria
   d. confer resistance to antibiotics

3. EcoRI cleaves DNA at
   a. AGGGTT b. GTATATC
   c. GAATTC d. TATAGC

4. Genetic engineering is
   a. making artificial genes.
   b. hybridization of DNA of one organism to that of the others.
   c. production of alcohol by using micro organisms.
   d. making artificial limbs, diagnostic instruments such as ECG, EEG etc.,

5. Consider the following statements:
   I. Recombinant DNA technology is popularly known as genetic engineering is a stream of biotechnology which deals with the manipulation of genetic materials by man invitro
   II. pBR322 is the first artificial cloning vector developed in 1977 by Boliver and Rodriguez from E.coli plasmid
   III. Restriction enzymes belongs to a class of enzymes called nucleases.

   Choose the correct option regarding above statements
   a. I & II b. I & III
   c. II & III d. I,II & III

6. The process of recombinant DNA technology has the following steps
   I. amplication of the gene
   II. Insertion of recombinant DNA into the host cells
   III. Cutting of DNA at specific location using restriction enzyme .
   IV. Isolation of genetic material (DNA)

   Pick out the correct sequence of step for recombinant DNA technology.
   a. II, III, IV, I b. IV, II, III, I
   c. I, II, III, IV d. IV, III, I, II

7. Which one of the following palindromic base sequence in DNA can be easily cut at about the middle by some particular restriction enzymes?
   a. 5` CGTTCG 3` 3` ATCGTA 5`
   b. 5` GATATG 3` 3` CTACTA 5`
   c. 5` GAATTC 3` 3` CTTAAG 5`
   d. 5` CACGTA 3` 3` CTCAGT 5`

8. pBR 322, BR stands for
   a. Plasmid Bacterial Recombination
   b. Plasmid Bacterial Replication
   c. Plasmid Boliver and Rodriguez
   d. Plasmid Baltimore and Rodriguez

9. Match the following :

Column A	Column B
1 Exonuclease	a. add or remove phosphate
2 Endonuclease	b. binding the DNA fragments
3 Alkaline Phosphatase	c. cut the DNA at terminus
4 Ligase	d. cut the DNA at middle

	1	2	3	4
A)	a	b	c	d
B)	c	d	b	a
C)	a	c	b	d
D)	c	d	a	b

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